

2.3 Functions & Features

- Multifunction: The APF is the solution to harmonics, inductive and capacitive reactive power as well as phase unbalance problem in a three-phase system.
- Superior harmonics compensation performance: The efficiency of harmonics compensation is up to 98%.
- Excellent reactive power compensation: The APF is capable of rapid (ms-grade), precise ($-0.99 < PF < 0.99$) and bi-directional (capacitive and inductive) reactive compensation.
- Wide input voltage and frequency range, suitable for the applications with diesel generators and harsh power supply conditions.

Module Type	Input Voltage Range	Input Frequency Range
Mixed fixed type 400V APF Module (3P4W)	228V ~ 456V	45 ~ 66 Hz
Mixed fixed type 400V APF Module (3P3W)	228V ~ 480V	45 ~ 66 Hz

- Automatic detection for the input frequency. The input frequency can be 50 Hz or 60 Hz.
- Auto start-up for the inverter: When the AC power source is back on, the inverter will be turned on automatically.
- Wide working temperature range

Module Type	Working Temperature Range with 100% Capacity	Capacity Derating over Temperature
Mixed fixed type 30A APF Module	-10°C ~ 50°C (14°F~122°F)	50% at 55°C (131°F) 0% at 55°C above
Mixed fixed type 50A APF Module	-10°C ~ 50°C (14°F~122°F)	50% at 55°C (131°F) 0% at 55°C above
Mixed fixed type 75A APF Module	-10°C ~ 40°C (14°F~104°F)	80% at 50°C (122°F) 50% at 55°C (131°F) 0% at 55°C above
Mixed fixed type 100A APF Module	-10°C ~ 50°C (14°F~122°F)	50% at 55°C (131°F) 0% at 55°C above
Mixed fixed type 150A APF Module	-10°C ~ 40°C (14°F~104°F)	80% at 50°C (122°F) 50% at 55°C (131°F) 0% at 55°C above